Problem Set: Five IV Designs

BUSS975: Causal Inference in Financial Research

Problem 1 (Lotteries: randomisation is not enough)

In 2008 Oregon expanded Medicaid for low-income adults. Facing excess demand, it allocated places by lottery: roughly 30,000 names were drawn from a list of 85,000. Of those selected, about 10,000 actually enrolled. Finkelstein et al. (2012) report:

Outcome	ITT (reduced form)	LATE (2SLS)
Has a usual place of care	9.9%	33.9%
Has a personal doctor	8.1%	28.0%
Got all needed healthcare	6.9%	23.9%
Any hospital admission	0.5%	2.1%
Had a medical collection	−1.6%	−6.4%
Owed on medical collection	−$99	−$390

(a) The lottery really was random. Explain why, even so, comparing those who *enrolled* in Medicaid with those who did not would be biased, and say in which direction. Name the selection at work.

(b) For each of the first three rows, back out the first-stage coefficient implied by the ITT and LATE columns. You should get roughly the same number three times. State what that number is, and what it means in words.

(c) Now do the same for the last three rows. You will get a somewhat different number. Both sets of estimates are correct as published. Give a reason why two different first-stage coefficients can both be right in the same experiment.

(d) Name the compliers, the always-takers and the never-takers in this design. Then explain why the LATE column says nothing about the always-takers, even though they are insured.

(e) A state legislator asks you which column to put in the briefing memo. A health economist asks you the same question. Give each of them an answer, and say why the answers differ.

Problem 2 (Leniency designs: judges and examiners)

A defendant i is randomly assigned to one of K bail judges within a court-and-period group. Let $D_i = \mathbf{1}\{\text{detained pre-trial}\}$, let $Y_i = \mathbf{1}\{\text{pleads guilty}\}$, and let Z_i be the vector of K judge dummies.

Parts (c)–(e) use the following nine cases. Judge A hears cases 1–4, judge B hears cases 5–7, and judge C hears cases 8–9. The detention outcomes are

$$D = (1,\ 1,\ 0,\ 0 \ \mid\ 1,\ 1,\ 1 \ \mid\ 1,\ 0).$$

(a) Write down the first and second stage, naming every object including the controls. Then say why the court-and-period controls are **not** optional, in one sentence.

(b) With K judge dummies as instruments, 2SLS is biased. Say what the bias is *toward*, and explain the mechanism in one sentence. Then explain why the following statement is wrong: *"the bias arises because each judge dummy is a weak instrument."*

(c) Write out the 9×3 matrix Z and the 3×3 matrix $Z'Z$. Say in words what the diagonal of $Z'Z$ contains, and why the matrix is diagonal.

(d) Compute the leverage $h_i = Z_i'(Z'Z)^{-1}Z_i$ for cases 1, 5 and 8. Then, for case 1, compute the leave-one-out leniency **two ways** — directly, as the mean detention rate of judge A's *other* cases, and from the formula $\hat{D}_i^{\text{JIVE}} = (Z_i'\hat{\pi} - h_i D_i)/(1 - h_i)$ — and show that they agree.

(e) Suppose a fourth judge heard exactly one case. What is that case's leverage, and what does the formula do? Say what this tells you to do in practice.

(f) JIVE leaves case i out of its own judge's detention rate. Explain why that is **not** enough once you add the court-and-period controls, and say what UJIVE does differently.

(g) Two courts assign cases differently. In court 1, each case is assigned to a judge one at a time, independently. In court 2, a single duty judge takes every case filed that day. State the correct standard errors in each court, and explain why clustering by judge is wrong in both.

(h) Goldsmith-Pinkham, Hull and Kolesár (2026) re-analyse 32,514 patent applications, instrumenting approval with examiner dummies and controlling for art unit $\times$ year. For the outcome "any subsequent application" they report UJIVE 0.173 (0.055), 2SLS with examiner dummies 0.232 (0.016), and OLS 0.234 (0.006). Two separate things have gone wrong in the 2SLS column. Name both, and explain why they are the *same* underlying problem showing up twice.

Problem 3 (Shift-share: the enclave instrument)

You want the effect of immigration on the wages of native workers:

$$Y_{l,t} \;=\; \alpha + \delta I_{l,t} + \rho X_{l,t} + \varepsilon_{l,t},$$

where l indexes a city, t a year, Y is log native wages, I is the immigrant inflow, and X holds city and year fixed effects. The instrument is $B_{l,t} = \sum_{k=1}^{K} z_{l,k,t_0}\, m_{k,t}$.

(a) Define z_{l,k,t_0} and $m_{k,t}$ in words, and say what k indexes. Then state the first and second stage, and name the unit of observation.

(b) Sign the bias in the OLS estimate of δ and explain the mechanism in one sentence. Then say what economic story would have to be true for OLS to be biased the *other* way.

(c) There are two separate defences of this instrument, and you need only one. State the identifying assumption of each in a single sentence, and name the diagnostic each one obliges you to run.

(d) Your data has 80 source countries, but the national surge from Mexico accounts for most of the variation in $m_{k,t}$. Which of the two defences survives and which does not? Explain.

(e) A referee writes: *"The 1980 enclave map is obviously endogenous — cities with large Mexican communities differ from other cities in a hundred ways."* Is the design dead? Answer precisely, and say what you would need to show to keep it alive.

(f) Design a pre-trend test for this instrument. Say exactly what you would regress on what, over which period, and what result would worry you.

Problem 4 (Fuzzy RD, and what a LATE is not)

(a) A fuzzy RD estimates

$$\delta_{\text{FRD}} = \frac{\lim_{x \downarrow c_0} E[Y|X=x] - \lim_{x \uparrow c_0} E[Y|X=x]}{\lim_{x \downarrow c_0} E[D|X=x] - \lim_{x \uparrow c_0} E[D|X=x]}.$$

Identify the numerator and the denominator by their IV names, and state the instrument.

(b) At a scholarship cutoff, crossing the threshold raises the probability of receiving aid by 0.25 and raises four-year graduation rates by 6 percentage points. Compute the fuzzy RD estimate and state precisely whose effect it is.

(c) The lecture note calls the fuzzy RD estimand "doubly local." Name the two localities.

(d) State what monotonicity means at an RD cutoff, and describe a concrete scholarship rule that would violate it.

(e) At the Fulton Fish Market the IV elasticity of demand is about -1.12, which looks elastic enough that a seller should cut prices and sell more. Explain why the seller should not do this, and then state the general lesson as a rule you could apply to any of the five designs.

Problem 5 (Computational: the elasticity of demand for fish)

Load `Fulton.dta` (`https://raw.github.com/scunning1975/mixtape/master/Fulton.dta`). The variables you need are `q` (log pounds of whiting sold), `p` (log average daily price), `Stormy` (bad weather at sea over the previous two days), and the day-of-week dummies `Mon`, `Tue`, `Wed`, `Thu`. There are 111 trading days. Control for the day-of-week dummies throughout and use heteroskedasticity-robust standard errors.

(a) Regress `q` on `p` and the day dummies. Report the coefficient and robust standard error. Why can this number not be read as the elasticity of demand?

(b) Estimate the first stage: regress `p` on `Stormy` and the day dummies. Report the coefficient, its robust standard error, and the effective F-statistic. Is the instrument strong? Interpret the coefficient in percentage terms.

(c) Estimate 2SLS, instrumenting `p` with `Stormy`. Report the coefficient and robust standard error. The IV estimate is roughly twice the OLS estimate in absolute value. Explain *why* OLS is the smaller of the two — your answer should mention what else is moving on a day when demand shifts.

(d) Compute the Anderson–Rubin 95% confidence interval by grid inversion. For a grid of candidate values δ_0, form $u_i = q_i - \delta_0 p_i$, regress u on `Stormy` and the day dummies, and keep those δ_0 for which the coefficient on `Stormy` is **not** significantly different from zero. Report the interval. Compare it with the conventional $\hat{\delta} \pm 1.96\,\widehat{\mathrm{SE}}$ interval, and say when the two would diverge sharply.

(e) The identifying claim is that a storm shifts supply but not demand. Give one concrete story under which a storm would also shift demand, and say what you would look at in the data to check it.